



Validity of primary driving tasks in head-mounted display-based driving simulators

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Abstract

The development of new car interior concepts requires tools, particularly in development phases before concept milestones, which enable subjective experiences and evaluations in static and driving situations. On the one hand, variant comparisons are necessary; on the other hand, the level of immersion should be high enough that participants can behave as they would in real cars. Virtual reality technologies and especially head-mounted displays are generally very suitable for such evaluations with the exception being in state-of-the-art driving simulators. Therefore, a validation study was undertaken in which primary driving tasks in two HMD-based simulators were compared with test runs in a real car. The difference between the simulators was only the state of the motion base (enabled vs. disabled). In both simulators and the test runs in the real car, four identical scenarios (straight, curves, overtaking and junction) were carried out. Since the focus is primarily on subjective ratings and gaze behaviour when evaluating new car interior concepts, in this study gaze behaviour was also priority. In addition, driving dynamics parameters were measured. The results reveal that the participants show more valid behaviour in the dynamic system than in the static simulator condition.

Keywords Driving simulation · Head-mounted display · Behavioural validity · Gaze behaviour · Eye tracking

1 Introduction

When developing new interior concepts for cars today, designers and engineers face multidimensional challenges like never before. In particular, the ongoing trend to add more car functions, the change to electric mobility and the development to highly automated and autonomous driving will change the appearance of interiors. For these reasons, there are many different things to be assessed in concept phases such as evaluations of visibilities, reachabilities, spatial feelings or dimensions and positions of concrete components like (head-up) displays or other control elements. Additionally, holistic concepts for the interaction of the driver with advanced driver assistance systems (ADAS), infotainment functionalities or mobile devices become more important. For this, the creation of a holistic brand image from opening, driving and closing the car will be an

essential competition advantage for the original equipment manufacturers (OEMs).

Driving simulators have become a central development tool during the recent decades. Advantages such as reproducible and fully parameterised scenarios, safe testing of critical driving situations, flexibility for complex variant comparisons, and the possibility to experience new car functions a long time before they are available in real prototypes have meant that these development tools are almost without alternative in concept development phases. But there are also some challenges when using driving simulators, such as the big variety of use cases with different requirements which prevents the use of one simulator for all investigations (Fischer et al. 2015; Hartfiel et al. 2018). Classical driving simulators are almost always equipped with displays or projection systems for visualisation and a mock-up as representation of the cockpit. Especially for the mock-up, engineers need to decide whether they want a realistic appearance to improve the immersion of the simulation or a generic version (e.g. consisting of item profile) to have more flexibility for positioning control elements. Both variants have drawbacks for the evaluation of new interior concepts. On the one hand, there is a realistic appearance, but there

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are limited options for replacing components. On the other hand, a generic mock-up is too abstract to create a feeling of sitting in a car. To support the development of new interiors, both advantages should be combined in a single tool. Virtual reality (VR) techniques offer some advantages. When using 3D-projection systems such as cave automatic virtual environments (CAVE) or head-mounted displays (HMD) for visualisation, all components can be virtual. Therefore, a simulation can be simultaneously highly immersive and flexible. In combination with some haptic dummies (e.g. for the steering wheel and displays), the driver–car interaction also becomes realistic. In some high-end driving simulators, CAVEs have been in use for years, but drawbacks like the expensive costs and the large amount of space required prevent a more frequent use also in smaller simulators. By contrast, HMDs with their restrictions regarding resolution, field of view (FOV) and the accommodation–vergence conflict were rather the exception in driving simulation context (Blissing and Bruzelius 2018). However, since new generations of HMDs were released in 2015–2016 (Oculus Rift, HTC Vive, Fove), this visualisation system is becoming an alternative for driving simulators. But because of the still existing drawbacks (e.g. the additional weight on the head) and other properties such as the complete decoupling from reality, influences with regard to the validity of driver behaviour must be taken into account.

2 HMDs in driving simulators

Blissing and Bruzelius (2018) investigated the suitability of HMDs for studies in the driving simulation context and reviewed 203 papers from the two largest conferences (Driving Simulation Conference, Road Safety and Simulation Conference). They summarised that only 4 studies

were conducted using a HMD-based simulator but also that 40% of all analysed studies could have been performed using an HMD. In addition, 47 papers (published between 1997 and 2019) in which HMDs were used as visualisation systems in a driving simulation context were investigated in the preparation of this publication. 24 of these papers were published between 1997 and 2015 and 23 between 2016 and 2019. 12 papers focused only theoretical/technical overviews or explanations without practical evaluations. In 35 papers, results from studies are reported. Four of them report research results for ergonomic investigations without driving. In 24 studies, very simple desktop simulators with game components or static simulators with mock-ups were used (Fig. 1).

7 studies were carried out in more complex simulators with motion bases. In two of these papers (Hirata et al. (2007), Ropelato et al. (2017), very small 2 and 6 DOF systems were used (e.g. with 65 mm range of motion). In two other papers, participants drove in systems with bigger motion bases, but the HMD visualisation worked only without head tracking (Erler and Rinderknecht 2017; Grabe et al. 2010). In the remaining three papers, driving simulators with bigger motion bases and head tracking were used (Hartfiel and Stark 2019; Tiemann 2019; Schill et al. 1997).

After analysing the studies, it could be summarised that 75% of the used simulators were static low or medium fidelity systems. Particularly for low budget simulators, the use of an HMD for visualisation is very attractive. Visual immersion levels can be achieved that could previously only be achieved with much more expensive systems like CAVEs or 3D projections. This has resulted in a much lower threshold for simulations of driving situations with high immersive visualisations. But it also increases the risk that simple and very fast built systems are used for investigations without previous validations.

Fig. 1 Two examples of used HMD-based driving simulators (Cao et al. 2019; Sullivan and Rothkopf 2012)



3 Transferability of results from the simulated to the real world

Simulators are normally used to investigate issues that would be too expensive, dangerous or take too long under real conditions. Therefore, the central goal for all simulators is the transferability of results to a real system. A quality criterion to describe the possibilities to transfer results from a simulated condition to a real one is the validity. Validity can be distinguished into physical and behavioural validity (Blaauw 1982; Mudd 1968). In the context of driving simulators, physical validity describes the physical correspondence between a simulated and a real driving environment. Physical validity can be improved, using a real car cockpit as mock-up or by visualising it in a virtual scene. It can also be improved if high-quality vehicle dynamic models are used in combination with motion bases.

Behavioural validity describes the correspondence between a simulated and a real driving scenario in the way that a participant behaves. Rolfe et al. (1970) describe it as: “The value of a simulator depends on its ability to elicit from the operator the same sort of response that he would make in the real situation”. Other researchers such as Blana (1996) and Zöllner et al. (2013) support this position and add that behavioural validity is in general more important than physical validity. Higher physical validity would be not enough, if the behaviour of the participant would be different to real conditions. On the other hand, behavioural validity can be reached with a simulator that does not have highest physical validity. In addition, behavioural validity can be differentiated into absolute and relative validity. Hirata et al. (2007) described these two criteria in this way: “Absolute validity is achieved if the numerical values between two systems are the same. [...] Relative validity is achieved if the differences found between experimental conditions are in the same direction, and have a similar or identical magnitude both real world and the simulator”.

Törnros (1998) reports that relative validity is required for a driving simulator. However, absolute validity is not essential. He argues that researchers often investigate effects of independent variables or differences between sample groups and not specific numerical values. But it isn't always easy to define borders from where control groups are similar and relative validity can be assumed. Therefore, the initial goal for all simulators stays to reach absolute validity.

Blaauw (1982) distinguishes between different methods to analyse behavioural validity. Besides the classical comparison between a simulated and a real condition, validity could be also investigated with comparing physiological parameters. Such comparisons are also common in the driving simulation context.

4 Validity of primary driving tasks in HMD-based driving simulators

This paper focuses on investigating the gaze behaviour in HMD-based driving simulators with the aim of evaluating the validity of primary driving tasks. Additionally, other parameters like physiological variables, speed-, acceleration- or steering- behaviour are also important for validations and have been taken in account.

4.1 Physiological parameters

Simulator sickness, immersion and presence are important indicators that realistic driver behaviour could be reached. Analysis of studies such as Aykent et al. (2014, 2015), Cao et al. (2019), Reich and Stark (2016), Ruspa et al. (2000) and Walch et al. (2017) comparing different visualisation systems revealed that HMDs led to higher levels of immersion and presence feeling but also higher simulator sickness scores than screens or 2D projection systems. The authors conclude that HMDs led generally to more simulator sickness scores compared to other visualisation systems without enough investigations about possible reasons. However, studies such as Weidner et al. (2017), Riecke et al. (2005) and Colombet et al. (2016) comparing HMDs with 3D-projection systems or CAVEs show instead comparable values for the parameters. Additionally, Hartfiel and Stark (2019) reveal that the increase of other cues (e.g. motion cues) led to a more homogeneous simulation and also lower simulator sickness scores. The same correlation can be seen in many studies for other visualisation systems. However, it could be argued that for HMDs just the risk for getting high simulator sickness scores is higher when the remaining simulator architecture is not appropriate for such high visual induced immersions. Therefore, the risk of getting invalid results from studies is also high, and comparisons to test runs where real cars were used are important. When looking through the studies analysed for this publication, it is noteworthy that only 2 of the 47 studies have done comparisons to real test runs.

4.2 Gaze behaviour

Since only one publication (Burns and Saluäär 1999) was found that contains validations of gaze behaviour parameters in an HMD-based driving simulator, studies such as Reich (2017) and Victor et al. (2005) on driving simulators with other visualisation systems were also analysed. Additionally, investigations of gaze behaviour in VR in other research contexts are taken into account Berton et al. (2019) and Brument et al. (2019).

Burns and Saluäär (1999) conducted a validation study in which they compare and analyse the visual behaviour of 11 participants in a static driving simulator with HMD visualisation and a real car. A t-junction is chosen as scenario. The results show that there is no difference in the number of glances, but drivers make significant longer fixations in VR than they did in the real test runs. Burns and Saluäär (1999) justify these findings with a need for longer glance times in VR to figure out whether cars are crossing the junction. Together with findings for other parameters, they summarise that driving in this HMD-based simulator is significantly different compared to driving in reality. One of the main reasons which they use for argumentation of the different gaze behaviour is the high proportion of simulator sickness (5 of 11 participants) during the simulator test runs. Although the results are already 21 years old, they are also relevant for these studies.

In the further explanations, studies that focus on gaze behaviour, but were carried out in driving simulators with other visualisation systems (e.g. screens or projection systems) were analysed.

In her doctoral thesis, Reich (2017) examined influences of visual and aurally parameters with regard to a valid driver behaviour. Comparisons of simulator variants with different immersion levels and test runs in a real car were performed. The task for the participants consisted of typing in addresses in a navigation system and perform Lane change tasks (LCT). The results show that participants looked less often to the navigation system in the high immersive variant. Reich (2017) explains these results with differences in the participants' sense of risk in the different test cases. She argues mainly that the participants in the low immersive simulator environment looked more often than in the real car longer than 1.6 s and 2 s to the navigation system. In the end, Reich concluded that the high immersive simulator variant is closer to real driving than the others and some parameters even show absolute validity.

Victor et al. (2005) carried out a study where they compared the performance of secondary tasks in different scenarios (motorway, rural roads) and driving simulators (two static simulators, one with a curved screen (135° FOV) the other one with a 2D projection system (150° FOV)). They also divide the tasks into different difficulty classes (Level 1–3). Although the focus of the study was not on the comparison between the real and the simulator environments, some findings were reported. The results show that the mean glance duration and the total glance duration are significant longer in the simulator than in the real world. They also reported significant differences in single difficulty levels for the parameters glance frequency and the percentage that participants looked to the road centre. These findings correlate with Reich's thesis that the sense of risk is lower in the low

fidelity simulators and therefore participants looked longer off the road (e.g. to navigation systems).

Because only one study Burns and Saluäär (1999) was found, where the gaze behaviour in a HMD-based driving simulator was investigated, additional studies about gaze behaviour in HMDs in other contexts are analysed.

Berton et al. (2019) investigated the gaze behaviour during collision avoidance between people in different virtual setups compared to the real one. The virtual environments were a Screen, HMD and a CAVE. Apart from the number of glances in the CAVE, the results show similar values for the other test environments. The authors argued the lower values in the CAVE with the lighting conditions which may have affected the eye tracking system. In general, they summarised that the values measured in the HMD come closest to the real ones.

Brument et al. (2019) carried out a study where participants should perform an eight-shaped trajectory using different strategies (walking, steering, beaming) in an HMD-based environment. Even if the results show similarities between the VR and real environments, they reported differences. They reported e.g. that the gaze anticipates more the change in heading in VR than in the real environment. The authors explained these results with three different points. First, the small FOV of the HMD could have an impact on the gaze anticipation and also the head movements. The second influence could be that the participants don't see their own body in VR and therefore they did more gaze anticipation. Third, the weight of the HMD could influence head movements and also gaze trajectories.

The previous analysis about the state of the art for eye tracking in the driving simulation context reveal on the one hand that there are a very limited number of investigations about the gaze behaviour in HMD-based driving simulators. Only one paper Burns and Saluäär (1999) was found, which was published over 20 years ago, where the technical possibilities were very limited. Therefore, findings may no longer stay true. On the other hand, studies in driving simulators with other visualisation systems show that valid gaze behaviour is in principle possible and could correlate with the level of presence of the whole simulation. In conclusion, the findings show that additional researches into the gaze behaviour of participants in relation to different driving simulator types are necessary.

4.3 Driving dynamics parameters

In order to examine the validity of primary driving tasks in HMD-based driving simulators, further parameters have to be considered. As mentioned in Sect. 4.1 the number of studies with comparisons between such simulators and real test runs is currently very limited (2 of 47 papers).

Burns and Saluäär (1999) not only reported results of the gaze behaviour, they also evaluated the validity of the speed behaviour in their validation study. Burns and Saluäär (1999) results show that in the t-junction scenario with the static HMD-based simulator, participants drove significantly slower compared to the real-world condition. They come to the conclusion that their simulator fidelity is not enough to create a valid driver behaviour.

Hirata et al. (2007) developed a compact HMD-based driving simulator with a very small 2 DOF (roll and pitch) motion base. They carried out a study to validate this device in urban underground expressway scenarios. During their study, Hirata et al. (2007) compared the measurements for speed, deceleration and distance behaviour. The results show that the participants tend to drive faster in the simulator environment for both speed levels (60 and 100 km/h), but there were no significant differences between both test conditions (chosen vs. required speed). For the acceleration behaviour, the authors compare the real-world condition with the simulator condition, split in a variant with activated motion base and one with deactivated motion base. The results show that the deceleration intensities are significantly higher in both simulator conditions. Additionally, this parameter shows significantly higher values for the static simulator compared to the dynamic version. For the distance behaviour, Hirata et al. (2007) reveal that the participants have chosen a smaller distance in the simulator than in the real-world test runs. However, most of these differences were not significant.

Grabe et al. (2010) compared different visualisation systems (screen or HMD without head tracking) in a dynamic driving simulator with regard to the drivers' performances without comparison to a real car. In the test runs, the participants should complete a slalom course and drive as smoothly as possible through the gates. The results show significant differences in the driving performance between both variants. In the HMD-based variant participants miss 27 gates in comparison with only one gate in the variant with the screen.

In conclusion, the discussed studies show that even if the simulator infrastructures are similar, the scenario and other simulation components could also have an influence on the driver behaviour. Investigators have to critically check, whether their simulators are appropriate for a specific research or not. However, in studies in HMD-based simulators, the proportion without validation is very high. For this, it is important to collect more experience using HMD-based driving simulators.

5 Methodology

In this paper, the results of a validation study are reported. During the investigations, test runs performed in two settings of an HMD-based driving simulator are compared to

the same test runs in a real car. The settings in the simulator only differ in the state of the motion base (disabled vs. enabled). The test runs are divided into 4 different scenarios:

- Straight,
- Curves,
- Overtaking and
- Junction.

The main focus of the validation was on gaze behaviour. This parameter is an important indicator to evaluate the driver behaviour, during an interaction with a new car interior concept in early development phases. Five different areas of interest (AOI) were defined to measure the attention ratio in different areas of the cockpit:

- Road scene ahead,
- Inside mirror,
- Instrument cluster (dashboard),
- Side window driver side (contains outside mirror driver side),
- Side window passenger side (contains outside mirror passenger side).

Percentage AOI attention ratio (ISO15007-1: 2014) is used as dependent variable. Because the goal for this study is to research the validity of primary driving tasks in these simulator environments, driving dynamics parameters are also measured. In this study, velocity, acceleration, deceleration and steering behaviour are chosen.

5.1 Hypotheses

Since driving simulators are often divided into static and dynamic systems in the literature, the main aim of this study is to validate these both types. Therefore, the following two main hypotheses are defined:

H1: There is no significant difference between the objective validity parameters (driver behaviour—and driving dynamics parameters) in the **static** HMD-based driving simulator compared to the test runs in a real car (absolute comparison between one specific simulator setup and test runs in a real car).

H2: There is no significant difference between the objective validity parameters (driver behaviour—and driving dynamics parameters) in the **dynamic** HMD-based driving simulator compared to the test runs in a real car (absolute comparison between one specific simulator setup and test runs in a real car).



Fig. 2 HMD-based driving simulator of the Volkswagen AG (Hartfiel et al. 2019)

5.2 Driving simulator, test car and measurement systems

For the test runs in the virtual environment the HMD-based driving simulator of the whole-vehicle development department of the Volkswagen AG was used (Fig. 2).

The simulator consists of an electro-mechanical hexapod. The cylinders have motion capacities of 300 mm. A variable seat buck is mounted on the motion base. In this seat buck, 16 linear motors enable the investigators to adjust the primary contact elements (steering wheel, pedals and seat) in the correct ergonomic positions for each car model. Due to the limited capacities of the motion base, an active seat belt loader is mounted on the seat with the goal to improve the feeling of deceleration during braking manoeuvres. This apparatus simulates braking forces relating to the brake pedal position. Head-phones with noise cancelling are used for the sound simulation consisting of engine-, wind-, tires- and traffic sound. For visualisation an HMD is used. During the test runs an HTC Vive Pro Eye with integrated eye tracking was used (State Autumn 2019). A detailed overview of the technical components of the simulator was published in the ATZ worldwide 2019 (Hartfiel et al. 2019).

To minimise bias due to different test environments the variable seat buck was adjusted to a Golf VII model in preparation and the vehicle dynamics model was customised to the parameters of the used test car (2.0 TDI with automatic transmission). Additionally, a diesel sound sample and the comparable cockpit visualisation was integrated in the simulator. For the test runs in the real world, a Golf VII with 2.0 TDI and automatic transmission was used (Fig. 3).

For the measurements in the real car, investigators used glasses from Ergoneers (Dikablis professional) for the eye tracking parameters. Additionally, a TTX-DataLogger from TTTech Group was connected with all Can-buses of the car to measure the driving dynamics parameters. In the



Fig. 3 Golf VII test car

simulator, a measurement tool developed from the company Cruden was used. This tool measured the data at 10 Hz.

5.3 Proving ground/scenario

The test runs in the real world were performed on a training area for novice drivers near Brunswick. There were some basic rules on the test ground. The participants had to maintain a top speed of 30 km/h. One other test vehicle (traffic car) on the test ground was required for the test runs with interaction between two traffic participants. The whole area consisted of different curve combinations, junctions and also straight roads with different traffic sign combinations.

The virtual scene was modelled as closely as possible to the real one. To this end, georeferenced data from the whole test ground were imported in a scenario editor tool. Based on these correct data, all streets were rebuilt with the original dimensions and all traffic signs were oriented in the correct positions. Additionally, native tree, bushes and other plants were positioned in a way that the whole scene looked very similar to its real counterpart (Fig. 4).

6 Study design

The study was divided into three parts, two simulator slots (static and dynamic) and one slot in the real car. Each participant drove once in each test environment (within subjects). The order was randomized to reduce influences of training effects in the results. Each slot was carried out on a different day to avoid influences of simulator sickness. The order of the different scenarios was not randomised. This was decided to reduce the proportion of incomplete data sets. Before the test runs started, a calibration of the eye tracking systems was carried out. Each slot included a 5 min familiarisation phase at the beginning. During this phase, each participant drove the same route to avoid initial influences

Fig. 4 Left: real test ground.
Right: virtual test ground



from different driving habits. After this, the four scenarios were carried out. The first was the straight road scenario that participants had to drive four times. The multiple execution was necessary, because of the limited dimensions of the test ground. The second scenario consisted of a defined trajectory with different sections including curves. One test run was about 2 min, and each participant drove this scenario twice. The third scenario was on the same section of the test ground as the straight road scenario. Participants were asked to overtake another traffic car that was driven by a second instructor. The traffic car was about 15 m in front of the test car and started to accelerate up to 20 km/h as soon as the test car started acceleration. In the fourth scenario, participants drove through a cross-junction and had to take a crossing traffic car into account. The crossing traffic car started to accelerate always on a specific position. This led to the situation in which participants always needed to wait the same amount of time and end up turning left. The scenario ended immediately after leaving the junction. Participants also drove this scenario twice.

6.1 Variables

As mentioned in Sect. 5, for the evaluation of the gaze behaviour the area of interest attention ratios (mean values and standard deviations) were calculated from the measured values of the eye tracking systems. This parameter was analysed for each AOI in each different scenario and each test environment. This parameter was also analysed for the complete overview (i.e. mean values summarised for all scenarios). Additionally, the following driving dynamics parameters were analysed for each scenario and also the complete overview (Table 1).

6.2 Participants

Certain requirements imposed during recruitment led to having only male participants, and VW employees. In general, more than 45 participants participated for at least one slot. Due to technical problems during the test runs, such as interruptions of the eyetracking system (e.g. due to a low sun) or the data logger, some data sets from the real car condition

Table 1 Driving dynamics parameters

Parameter	Analysed values	Unit
Velocity	Mean maximum velocity	km/h
	Mean velocity	km/h
Acceleration/deceleration	Mean maximum acceleration	m/s ²
	Mean minimum deceleration	m/s ²
Steering behaviour	Mean maximum steering angle	°
	Number of changes of directions	

Table 2 Age distribution in the scenarios and in total

Age Group/Scenario	25–39	40–49	50–55	Older than 55	Total (all slots in a scenario)
Straight	3	11	2	4	20
Curves	3	9	1	4	17
Over-taking	3	8	1	3	15
Junction	3	8	1	3	15
Complete	3	8	0	3	14

were not complete. There were also incomplete data sets in the simulator. Reasons for this were simulator sickness symptoms and also interruptions in the measurement due to technical problems. Additionally, there also were other reasons such as illnesses that caused participants to not be able to participate the slots. This led to incomplete data sets. Table 2 visualises the distributions of participants in different scenarios (i.e. complete data sets for all slots in a specific scenario) and the complete overview (i.e. complete data sets for all scenarios in all slots) and the age groups of participants.

7 Results

Different methods were used for analysis. The statistical approach is based on Zöller (2015). The level of significance is $\alpha = 0.05$, if not reported otherwise. In this study, a within subjects design is used, and there are dependent samples.

At first all samples are tested for normal distribution. If there was normal distribution, one-factor repeated-measures analysis of variance (ANOVA) was carried out. For this, Mauchly tests for sphericity were performed. This test checked the homogeneity of the correlations between the different factor levels. If there was a significant result, then the correlations of the samples would not be homogeneous. In these cases, Greenhouse–Geisser-corrections were conducted. If there was sphericity, test for innersubject effects could be carried out without corrections. During the tests of the innersubject effects, repeated-measures ANOVAs were calculated. If the repeated-measures factors are $p \leq 0.05$, the influence of the parameter is significant. In these cases, post hoc analysis between different levels (static simulator, dynamic simulator and real car) of the factor were performed. For this purpose, pairwise comparisons of mean values (t test for dependent variables) with Bonferroni correction were carried out in order to investigate which mean values differ significantly. If the repeated-measures ANOVA shows no significant difference between the three conditions, it was assumed that there is absolute validity for these values in the simulator environments.

If the test for normal distribution was negative, repeated-measures ANOVA was not possible. In these cases, Friedman's ANOVA as a nonparametric test was carried out. If the Friedman tests were significant ($p \leq 0.05$), this was a sign that there is an influence of the repeated-measures factor. In the next step, a post-hoc analysis was also performed. Since there were no normal distributions, the Wilcoxon rank-sum test was used. For these tests, Bonferroni corrections were also considered. Because of these corrections, the initial level of significance ($\alpha = 0.05$) value must be divided by the number of measurement repeats ($k = 3$). This led to a level of significance of $\alpha = 0.0167$. P-values where this adjusted level of significance was used, are marked by an "a".

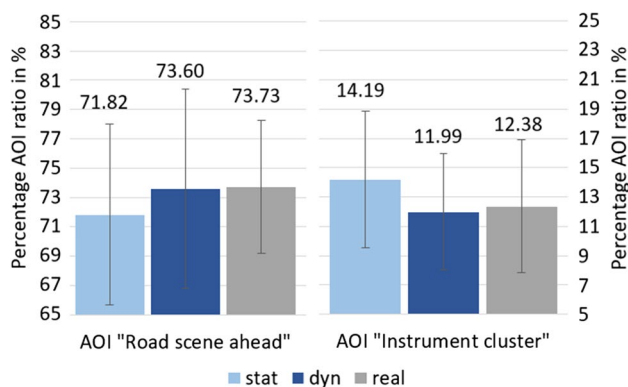


Fig. 5 Percentage AOI ratio for AOI: Road scene ahead (left) and Instrument cluster (right) (all scenarios)

7.1 Gaze behaviour

In the following, results for the entire study will be presented first followed by the ones for the single scenarios. The bar charts (Figs. 5, 6, 7, 8 and 9) show the summarised results for the entire study for the different AOIs. The repeated-measures ANOVA (sphericity assumed: Mauchly-W(2)=0.957, $p = 0.767$) for the AOI "Road Scene ahead" shows no significant differences between the three test conditions ($F(2,26) = 0.62$, $p = 0.54$, partial $\eta^2 = 0.05$, $n = 14$). Nevertheless, mean values show that participants looked less often on the street in the static simulator (Fig. 5).

The repeated-measures ANOVA (sphericity assumed: Mauchly-W(2)=0.871, $p = 0.436$) for the AOI "Instrument cluster" reveals that there are no significant differences between the conditions ($F(2, 26) = 1.26$, $p = 0.30$, partial $\eta^2 = 0.09$, $n = 14$). For this AOI, mean values show that participants look more often at the instrument cluster in the static simulator (Fig. 5).

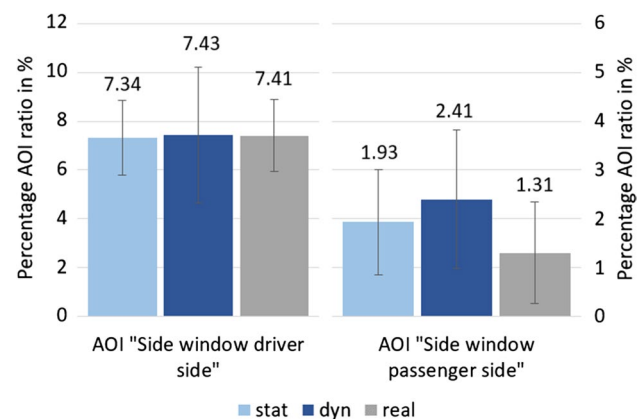


Fig. 6 Percentage AOI ratio for AOIs: Left: Side window driver side, Right: Side window passenger side (all scenarios)

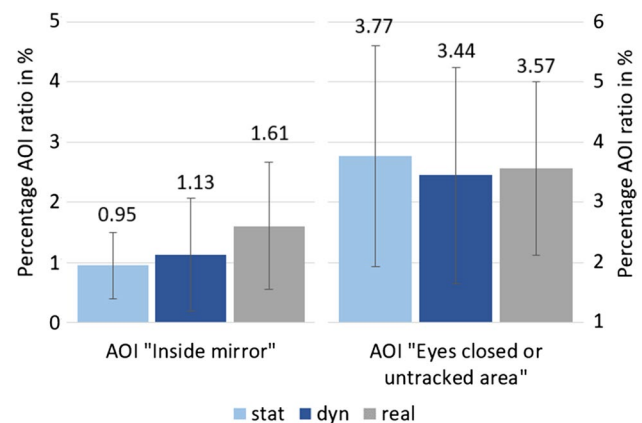


Fig. 7 Percentage AOI ratio for AOIs: Left: Inside mirror, Right: Eyes closed or untracked area (all scenarios)

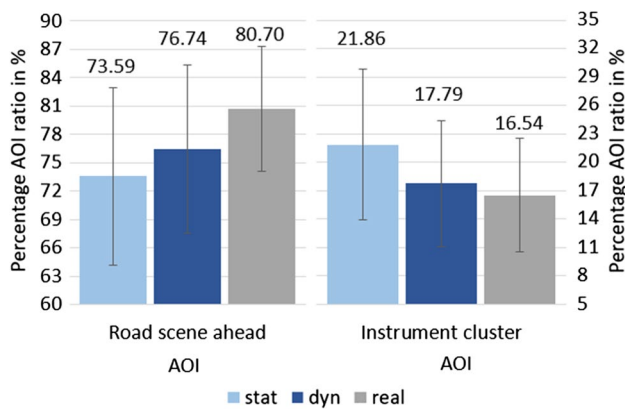


Fig. 8 Percentage AOI ratio for AOIs: Left: Road scene ahead, Right: Instrument cluster (scenario “Straight”)

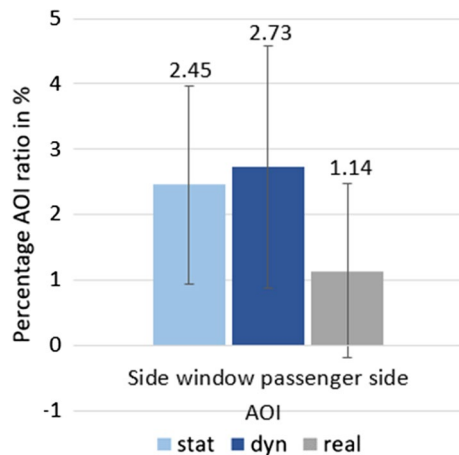


Fig. 9 Percentage AOI ratio for AOI: Side window passenger side (scenario “Curves”)

For the AOI “Side window driver side”, Mauchly-test for sphericity is significant ($W(2)=0.595, p=0.044$). For this, repeated-measures ANOVA with Greenhouse-Geisser-correction was carried out ($F(1.423, 18.503)=0.01, p=0.969$, partial $\eta^2=0.001, n=14$). The results show no significant difference between the three test conditions (Fig. 6). The results for the AOI “Side window passenger side” show no significant differences. As these values were not normally distributed, Friedman’s ANOVA was performed ($\chi^2(2)=5.571, p=0.062, n=14$). The mean values for the driver side are almost the same, whereas the values for the passenger side reveal some differences (Fig. 6).

For the AOI “Inside mirror”, the repeated-measures ANOVA (sphericity assumed: Mauchly- $W(2)=0.639, p=0.068$) shows that there are no significant differences between the three test conditions ($F(2, 26)=1.991, p=0.157$, partial $\eta^2=0.133, n=14$). The mean values

show that participants look less often through the inside mirror in both simulator conditions (Fig. 7).

During the test runs, there were times, when the participants glanced or looked at an area in which no AOI was defined, e.g. the dashboard in the middle of the cockpit. The results for this “AOI”-“Eyes closed or untracked area” show similar values, and the repeated-measures ANOVA (sphericity assumed: Mauchly- $W(2)=0.659, p=0.082$) also reports no significant difference between the conditions ($F(2, 26)=0.21, p=0.81$, partial $\eta^2=0.02, n=14$). The mean values also show similar values (Fig. 7).

Since results for most of the values in the different scenarios also show no significant differences, only cases where significant differences occurred would be reported. In the scenario “Straight”, the results show significant differences for AOIs “Road scene ahead” and “Instrument cluster”. For the AOI “Road scene ahead”, the repeated-measures ANOVA (sphericity assumed: Mauchly- $W(2)=0.973, p=0.783$) reveals that there are significant differences between the conditions ($F(2, 38)=5.256, p=0.01$, partial $\eta^2=0.217, n=20$). Therefore, post-hoc t-tests for dependent samples were carried out. These tests show that the static simulator generates a significant different ratio for this AOI ($t(19)=-3.386, p=0.003$). On the other hand, the dynamic simulator generates no significant different ratios ($t(19)=-1.917, p=0.07$). The mean values also show these differences (Fig. 8).

For the AOI “Instrument cluster”, in the scenario “Straight”, the repeated-measures ANOVA (sphericity assumed: Mauchly- $W(2)=0.999, p=0.988$) shows significant differences ($F(2, 38)=3.448, p=0.042$, partial $\eta^2=0.154, n=20$). The post hoc t-tests show that these differences are between the static simulator condition and the real car test runs ($t(19)=2.492, p=0.022$). For the dynamic simulator condition, the results show no significant differences ($t(19)=0.537, p=0.597$). The mean values support these findings (Fig. 8). The repeated-measures ANOVA (sphericity assumed: Mauchly- $W(2)=0.954, p=0.656$) for the AOI “Eyes closed or untracked area” in the scenario “Straight” reveals also significant differences ($F(2, 38)=8.364, p=0.001$, partial $\eta^2=0.306, n=20$). The post-hoc t-tests also show these results for both simulator conditions compared to the test runs in the real car (stat: $t(19)=3.347, p=0.003$; dyn: $t(19)=3.408, p=0.003$). The mean values support these results. In the scenario “Straight”, the other AOIs show no further significant differences.

In the results of the scenario “Curves”, only the values for the AOI “Side window passenger side” show significant differences. For this AOI, the repeated-measures ANOVA (sphericity assumed: Mauchly- $W(2)=0.759, p=0.127$) shows significant differences between the two simulator conditions compared to the test runs in the real car ($F(2, 32)=7.424, p=0.002$, partial $\eta^2=0.317, n=17$). Post-hoc

t-tests confirm the findings (stat: $t(16)=2.976$, $p=0.009$; dyn: $t(16)=3.031$, $p=0.008$) and also mean values (Fig. 9).

As the “Inside mirror” AOI was not normally distributed, Friedman’s ANOVA was carried out and became significant ($\chi^2(2)=6.000$, $p=0.05$, $n=17$). But the post-hoc Wilcoxon-tests show no significant difference for both simulator conditions (stat: $z=-2.154$, $p=0.031^a$, dyn: $z=-1.603$, $p=0.109^a$). Nevertheless, mean values reveal that (in %) participants looked less often through this AOI in the real car (stat: $M=1.10$, $SD=3.89$; dyn: $M=0.78$, $SD=1.25$, real: $M=1.75$, $SD=2.01$).

The results for the scenario “overtaking” show no significant differences for all AOIs except for the AOI “Inside mirror”. For this AOI, repeated-measures ANOVA (sphericity assumed: Mauchly-W(2)=0.635, $p=0.052$) shows significant differences between the test conditions ($F(2, 28)=4.738$, $p=0.017$, partial $\eta^2=0.253$, $n=15$). Post-hoc t-tests show that the values for the static simulator are significantly different compared to the real-world test runs ($t(14)=2.864$, $p=0.013$). However, the pairwise t-test between the dynamic simulator and the real car condition reveals no significant difference ($t(14)=2.044$, $p=0.065$). The mean values show that (in %) participants look more often in this AOI in both simulator conditions (stat: $M=4.49$, $SD=1.79$; dyn: $M=4.08$, $SD=2.73$; real: $M=2.55$, $SD=1.72$).

The results for the scenario “Junction” show significant differences between the static simulator condition and the real car test runs for one AOI and values close to significance for additional one AOI. As the “Inside mirror” AOI was not normally distributed, Friedman’s ANOVA was performed ($\chi^2(2)=5.907$, $p=0.052$, $n=15$). Statistical tests that have to work with unequal variances are generally less precise. Therefore, in this case post-hoc Wilcoxon-tests were carried out because the p-value is very close to significance. These tests show significant differences between the static simulator condition in the real car ($z=-2.667$, $p=0.008^a$). The value for the dynamic condition is not significant because of the adjustment of the level of significance ($z=-2.197$, $p=0.028^a$). Mean values show differences, too ((in %) stat: $M=0.30$, $SD=0.52$; dyn: $M=0.43$, $SD=0.54$; real: $M=2.68$, $SD=4.07$). For the AOI “Side window passenger side”, Friedman’s ANOVA was performed, because the values are not normally distributed ($\chi^2(2)=7.356$, $p=0.025$, $n=15$). For this AOI, post-hoc Wilcoxon-tests reveal no significant difference for both simulator conditions (stat: $z=-1.306$, $p=0.191^a$; dyn: $z=-1.590$, $p=0.112^a$).

7.2 Driving dynamics parameters

For the driving dynamics parameters, results from the entire study for all scenarios are reported followed by the results for each scenario. Different from the gaze behaviour parameters,

for the driving dynamics parameters most analysis reveal significant differences between the simulator conditions and the real-world test runs. Particularly for the parameters “mean max. acceleration” and “mean max. deceleration”, significant differences were determined for both simulator variants in all scenarios. Therefore, only these results are reported, where at least one simulator condition shows no significant differences in comparison with the real test runs.

For the entire study, the values for the parameter “Velocity” (mean max. and mean) show no significant differences between the simulator conditions compared to the real test runs. For the mean max. velocity, a Friedman’s ANOVA was performed and became not significant ($\chi^2(2)=1.857$, $p=0.395$, $n=14$). The repeated-measures ANOVA (sphericity assumed: Mauchly-W(2)=0.799, $p=0.260$) for the parameter “mean velocity” also shows no significant differences between the three test conditions. Nevertheless, the results for both parameters show that the participants drove faster in the static simulator in general (Fig. 10).

The results for all other driving dynamics parameters reveal significant differences between the simulator conditions and the real car test runs for the entire study analysis.

In the scenario “Straight”, the results show that the mean max. velocity is not significantly different in both simulator conditions in comparison with the real test runs. The Friedman’s ANOVA did not become significant ($\chi^2(2)=1.848$, $p=0.397$, $n=20$). Mean values are visible in Fig. 11. For the parameter “mean velocity”, the repeated-measures ANOVA (sphericity assumed: Mauchly-W(2)=0.860, $p=0.256$) shows significant differences ($F(2, 38)=6.737$, $p=0.003$, partial $\eta^2=0.262$, $n=20$). The post-hoc t-tests are only significant for the static simulator (stat: $t(19)=3.340$, $p=0.003$; dyn: $t(19)=2.101$, $p=0.053$). The mean values also show that participants drove faster in the static simulator (Fig. 11).

For the parameter “mean max. steering angle”, Friedman’s ANOVA reveals no significant difference (χ^2

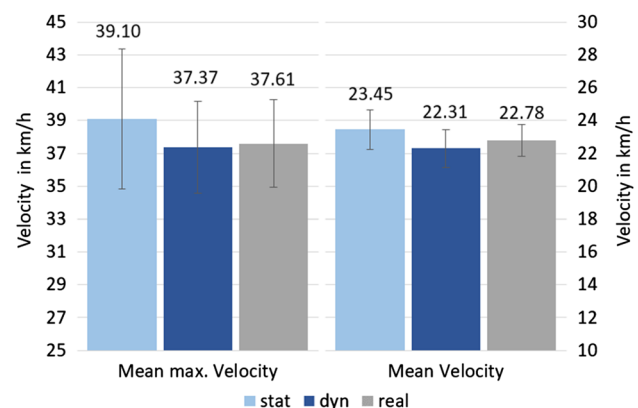


Fig. 10 Left: Mean max. velocity. Right: Mean velocity (all scenarios)

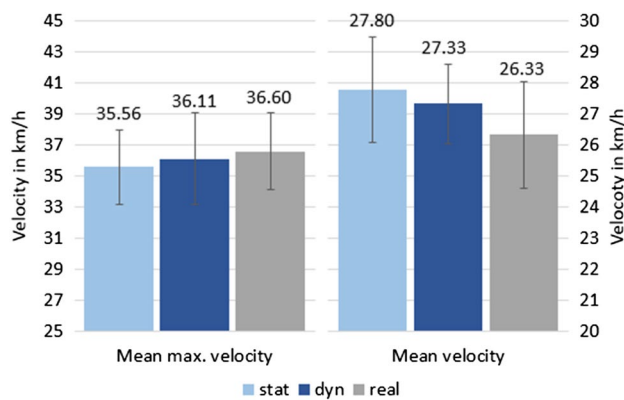


Fig. 11 Left: Mean max. velocity. Right: Mean velocity (Scenario “Straight”)

(2) = 1.300, $p = 0.522$, $n = 20$). The mean values are also very similar: (in km/h) stat: $M = 41.19$, $SD = 14.44$; dyn: $M = 41.21$, $SD = 7.90$; real: $M = 40.98$, $SD = 4.64$.

In the scenario “Curves”, the results for both simulator conditions show no significant differences for the parameter “mean max. velocity” ($\chi^2(2) = 4.235$, $p = 0.120$, $n = 17$) (Fig. 12). For the parameter “mean velocity”, there was also no normal distribution achieved. The Friedman’s ANOVA became significant ($\chi^2(2) = 9.529$, $p = 0.009$, $n = 17$). The post-hoc Wilcoxon-tests show for both simulator conditions compared to the real test runs no significant differences (stat: $z = -1.207$, $p = 0.227^a$; dyn: $z = -1.586$, $p = 0.113^a$). However, the Wilcoxon-test between the two simulator conditions becomes significant ($z = -2.675$, $p = 0.007^a$). The mean values show that for this parameter in this scenario the participants drove a little bit faster in the static simulator (Fig. 12).

All other parameters reveal significant differences for both simulator conditions compared to the real test runs in this scenario.

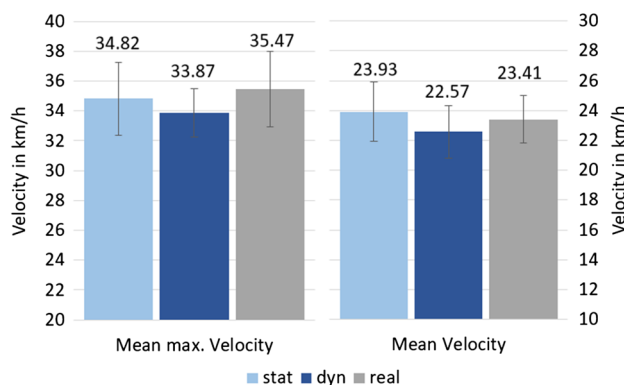


Fig. 12 Left: Max. velocity. Right: Mean velocity (Scenario “Curves”)

In the scenario “Overtaking”, for the parameter “mean max. velocity” the repeated-measures ANOVA (sphericity cannot be assumed Mauchly-W(2) = 0.626, $p = 0.047$) with Greenhouse–Geisser-correction shows no significant differences for the two simulator conditions ($F(1.455, 20.373) = 2.854$, $p = 0.094$, partial $\eta^2 = 0.169$, $n = 15$). The mean values show that (in km/h) participants drove a little faster in the static simulator (stat: $M = 38.36$, $SD = 4.53$; dyn: $M = 35.74$, $SD = 2.41$; real: $M = 35.66$, $SD = 2.44$). For the parameter “mean velocity”, the repeated-measures ANOVA (sphericity assumed: Mauchly-W(2) = 0.983, $p = 0.893$) became significant ($F(2, 28) = 10.316$, $p = 0.000$, partial $\eta^2 = 0.424$, $n = 15$). The results for the post-hoc t-tests show that only the values for the static simulator conditions differ significantly from them for the real test runs (stat: $t(14) = 4.268$, $p = 0.001$; dyn: $t(14) = 1.762$, $p = 0.100$). Mean values also show that (in km/h) participants drove faster in the static simulator (Fig. 13). The results for the parameter “mean max. steering angle” show no normal distribution. The Friedman’s ANOVA became not significant ($\chi^2(2) = 5.733$, $p = 0.057$, $n = 15$). Mean values show that (in °) the max. steering angle is bigger in the static simulator than in the dynamic and the real test runs (Fig. 13). All other parameters reveal significant differences for both simulator conditions.

In the scenario “Junction” for the parameter “mean max. velocity”, the repeated-measures ANOVA (sphericity assumed: Mauchly-W(2) = 0.969, $p = 0.813$) became significant ($F(2, 28) = 3.887$, $p = 0.032$, partial $\eta^2 = 0.217$, $n = 15$). The post-hoc t-tests show significant differences for the values of the static simulator ($t(14) = 2.629$, $p = 0.020$) but not for the dynamic simulator ($t(14) = 1.481$, $p = 0.161$). Mean values also show that participants drove faster in the static simulator (Fig. 14). For the parameter “number of changes of directions”, the repeated-measures ANOVA (sphericity assumed: Mauchly-W(2) = 0.999, $p = 0.994$) shows no significant differences ($F(2, 28) = 1.590$, $p = 0.222$, partial

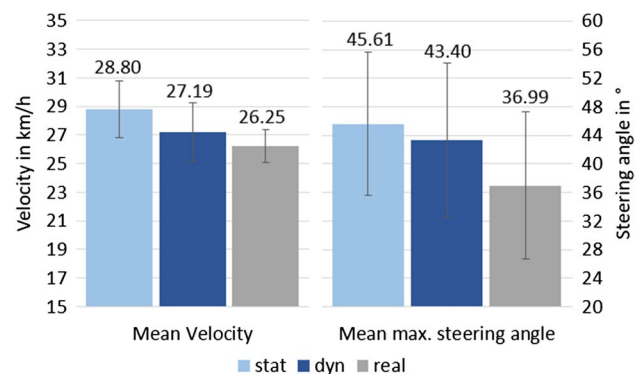


Fig. 13 Left: Mean velocity. Right: Mean max. steering angle (Scenario “Overtaking”)

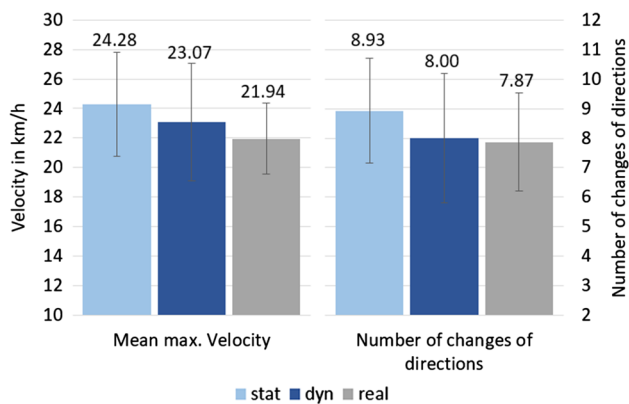


Fig. 14 Left: Mean max. velocity. Right: Number of changes of directions (scenario “Junction”)

$\eta^2 = 0.102$, $n = 15$). Nevertheless, the mean values show that participants change the steering direction a little more often in the static simulator conditions (Fig. 14). For the other parameters, the results reveal significant differences between both simulator conditions compared to the real-world test runs in this scenario.

8 Discussion

8.1 Gaze behaviour

For the entire study, results show no significant differences for both simulator conditions compared to the real-world test runs. That means that these two HMD-based simulator variants generate absolute valid gaze behaviour with regard to the parameter “percentage attention ratio”. Therefore, both hypotheses can be maintained. A closer look at the different mean values for the entire study show that participants looked less often at the road ahead and more often at the instrument cluster in the static simulator, compared to the dynamic simulator and the real-world test runs. Values suggest that in the static simulator for the participants it could have been more difficult to estimate the current velocity. However, findings could additionally mean that participants do not have the risk awareness that they would have in the real car. The values for the dynamic simulator are in both cases closer to real conditions. This allows the claim that speed feeling and also risk awareness are more realistic in the dynamic simulator condition.

Results of the scenario “Straight” support the general findings. In this scenario, the difference for the parameter “percentage attention ratio” becomes significant between the static simulator and the real-world condition for the AOIs “road scene ahead” and “instrument cluster”. In addition, for both simulator conditions, values for the AOI

“eyes closed or untracked area” show significant differences in comparison with the real test runs. A possible reason for these differences could be the use of different measurement systems or the not completely identical AOIs. Although the AOIs are the same in the simulator conditions, in the real car they have to be redefined. For the static simulator, this means that the hypothesis must be rejected because of the significant differences for 3 AOIs. But the differences are small enough for relative validity to be assumed. The values for the AOI “Eyes closed or untracked area” are not a directly measured parameter. Therefore, it can be summarised that the dynamic simulator condition generates absolute validity for this scenario because no other AOI shows significant differences.

In the scenario “Curves”, for all AOIs except the AOI “side window passenger side”, the values show absolute validity for both simulator conditions. For the AOI “side window passenger side” the values reveal significant differences between the two simulator conditions compared to the real test runs. The reason could be that in the real car, when taking sharp curves, participants moved their eyes more than in the simulator conditions. Therefore, sometimes the eye tracking system could not track the pupils. This thesis is supported by the results for the AOI “Eyes closed or untracked area”, which contains these durations. In this scenario, the values are higher in the real car compared to the simulator conditions. In addition, the reduced horizontal FOV of the HMD (110°) could cause the participants to have to turn their heads more in the simulator to look at the curves. As a result, it’s possible that they look more often through the side window. Therefore, both hypotheses cannot be maintained for this scenario. But because only one AOI show small differences between the two simulator conditions and the real car, it can be assumed that both simulator conditions show relative validity.

In the scenario “Overtaking”, all AOIs except the AOI “Inside mirror” show absolute validity. The values for this AOI show that participants looked significantly more often in the inside mirror in the static simulator condition compared to the real test runs. A reason for these differences could again be the reduced FOV and also the low resolution of the HMD, which could have led to the participants having difficulty seeing the other vehicle in the mirror. In addition, the mirrors in the simulator conditions support no movement parallax during the study. Based on the findings, it can be concluded that the static simulator generates only relative validity in this scenario. However, the results for the dynamic simulator show absolute validity.

In the scenario “Junction”, for all AOIs except the AOI “Inside mirror” results show no significant difference. For the AOI “Inside mirror”, only the static simulator condition shows significant differences. Therefore, it can be argued

that the static simulator generates only relative validity. However, in the dynamic simulator participants show absolute valid gaze behaviour with regard to the parameter “percentage attention ratio”.

8.2 Driving dynamics parameters

For the analysis for the entire study, the results reveal that values for the parameters “mean max. acceleration” and “mean max. deceleration” are significantly different in both simulator conditions in comparison with the real-world test runs. These findings are also confirmed in the four scenarios. On the one hand, the reason could be that in both simulator conditions the motion cues are not intensive enough that the participants feel the acceleration. But the results for the parameters show no systematic differences. In the scenarios “Straight” and “Curves”, the dynamic simulator shows higher mean max. values, whereas in the scenarios “Overtaking” and “Junction” the static simulator generates higher values. Therefore, it can be also argued that the car model in the simulator is not good enough at the moment. The results for the steering parameters (mean max. steering angle and number of changes of directions) also reveal significant differences for the general overview (all scenarios). Because the differences were comparably big, validity cannot be assumed for both simulators for these four parameters. For the parameters “mean max. velocity” and “mean velocity”, the results show in general (all scenarios) no significant differences and therefore it can be assumed that both simulator conditions show absolute validity with regard to these mean values.

In the scenario “Straight”, the results reveal that the values for the parameter “mean max. velocity” show no significant differences between the simulator conditions and the real-world test runs. For the parameter “mean velocity”, the results show that only the dynamic condition shows absolute validity, whereas in the static simulator participants drove significantly faster. These findings support the results from the gaze behaviour, where participants also had difficulties maintaining the correct speed and looked more often to the instrument cluster. The results for the parameter “mean max. steering angle” reveal no significant differences, whereas the parameter “number of changes of directions” shows significant differences for both simulator conditions. Maybe for participants it is more difficult to stabilize the car in the simulator conditions than in the real car. Reason can be again the difference between the virtual car model compared to the real car.

In the scenario “Curves”, only the parameters “mean max. velocity” and “mean velocity” show no significant differences for the simulator conditions. Therefore, absolute validity can be also assumed only for these two parameter for both simulator conditions. The differences for the

acceleration and deceleration values are too big to assume relative validity. However, for the parameters “mean max. steering angle” and “number of changes of directions” the differences are limited. Therefore, for these parameters relative validity can be assumed.

In the scenario “Overtaking”, the results show that the static simulator generates only for the parameter “mean max. velocity” and “mean max. steering angle” absolute validity, whereas for all other parameters significant differences occurred. For the parameters “mean velocity” and “number of changes of directions”, these differences are small enough to assume relative validity. But for the parameters “mean max. acceleration” and “mean max. deceleration”, the differences are too big, which is why no validity can be assumed. The values for the dynamic simulator reveal absolute validity for the parameters “mean max. velocity”, “mean velocity” and “mean max. steering angle”. In particular, the results for the velocity parameters support the findings that with the additional motion cues it is easier for participants to hold the correct velocity. The results for the parameters “mean max. acceleration”, “mean max. deceleration” show big differences as to why validity cannot be assumed. For the parameter “number of changes of directions”, the differences between the simulator conditions and the real test runs in the real car are small enough that relative validity can be assumed.

In the scenario “Junction” only for the parameter “number of changes of directions”, the results show absolute validity for both simulator variants. For the parameter “mean max. velocity”, only the dynamic simulator shows absolute validity, whereas the static condition generates relative validity of driver behaviour. For the parameter “mean velocity”, the results reveal that the both simulator conditions generate relative validity. The values show that the dynamic condition is closer to the real-world condition. These findings also support the general findings of this study. For the parameters “mean max. acceleration”, “mean max. deceleration” and “mean max. steering angle”, the results show no validity. This also supports the findings that probably the parameters of the car model were not close enough to the real car.

9 Conclusion

In this study, the main goal was to evaluate the capabilities to generate realistic driver behaviour in two different HMD-based driving simulators compared to tests runs in a real car. The use case behind these investigations is the subjective evaluation of new car interior concepts during driving situations in early development phases. The results show that in general, for the chosen gaze behaviour parameter, both simulator conditions generate at least relative validity for all AOIs in all scenarios. However, it is important to note

that in most cases the values for the dynamic simulator are closer to the real-world conditions. These findings are also confirmed by the statistical analysis. It becomes obvious that in the static simulator the values for 6 of 30 AOIs reveal significant differences and in the dynamic simulator only 2 of 30 show significantly different mean values. Additionally, the differences in the static simulator occur in central AOIs like “road scene ahead” and “instrument cluster”. For this, especially for the very subjective evaluations of car interior concepts, it is recommended to expand a virtual environment with motion cues as soon as evaluations are necessary during driving.

In addition, driving dynamics parameters (velocity, acceleration, deceleration, steering behaviour) are analysed in this study. For the two velocity parameters, the results show that in 4 of 10 measured cases the values for the static simulator show significant differences, whereas in the dynamic simulator only 1 of 10 cases show such different values. Together with the findings for the gaze behaviour, these results suggest that participants had difficulties holding a given velocity in the static condition, additionally showing less risk awareness. Both results are indicators that participants accepted the dynamic simulation more as driving in a car than a static simulation. In addition, the results reveal the importance of correct car models for the validity of driver behaviour. Most of the driving dynamics values for the parameters acceleration, deceleration, and steering behaviour reveal no validity (17 of 30 parameters in both simulator variants). Additionally, for these parameters the results suggest that the motion cues can have a positive influence with regard to more realistic driving behaviour.

For this study, only rough AOIs were chosen because the primary goal was the basic validation of driving in HMD-based simulators. Next steps should be comparable evaluations for more detailed and concrete use cases such as interaction with touch screens to find out more about the possibilities of HMD-based driving simulators.

Compliance with ethical standards

Conflict of interest All the authors declare that they have no conflict of interest.

Ethical approval The whole study design was approved by the ethical and data protection board of the Volkswagen AG and also in constant coordination with the Technical University of Berlin as independent organisation. Before each session in the driving simulators or the real car, participants signed declaration of consent and got a security safety briefing.

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